



FEDERAL UNIVERSITY LAFIA
FACULTY OF SCIENCE
DEPARTMENT OF CHEMISTRY

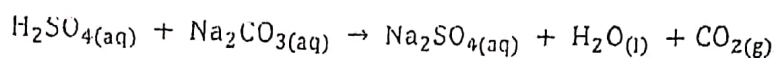
First Semester Examination for 2017/2018 Session
CHM 112 – Practical Chemistry I
Duration: 1h

Credit Unit: 1

Instruction: Attempt All Questions

Question 1

Solution B contains 1.325 g of anhydrous Na_2CO_3 per 250 cm^3 of solution. Solution A (H_2SO_4) was titrated against 25 cm^3 portion of solution B using methyl orange as an indicator. The equation for the reaction involved in the titration is;



The following results obtained are shown in the Table below

Titration number	Titre value (cm^3)		
	1	2	3
Final burette reading	28.60	41.80	18.70
Initial burette reading	15.50	28.30	5.50
Volume of acid used	13.10	$X = 13.5$	13.20

From the results and the information provided, calculate;

- (a)
- (i) The titer value, X , for the second titration.
 - (ii) The average volume of acid used.
- (b) The concentration of Na_2CO_3 in mol/dm^3
- (c) The number of moles of Na_2CO_3 in volume used
- (d) The equivalent mole of H_2SO_4 used.
- (e)
- (i) The concentration of H_2SO_4 in mol/dm^3
 - (ii) The concentration of H_2SO_4 in g/dm^3

($\text{H}=1$, $\text{C}=12$, $\text{O}=16$, $\text{S}=32$, $\text{Na}=23$)

- (f) State three (3) precautions necessary to obtain accurate result in the experiment.
- (g) Name the apparatus required for accurate measurement of solution B.

$$\begin{aligned} 1.325 \text{ g} &= 250 \text{ cm}^3 \\ \text{Na}_2\text{CO}_3 &= 106 \text{ g/mol} \\ \therefore X &= \frac{13.25}{250} = 5.3 \text{ g/dm}^3 \\ \text{conc} &= \frac{5.3}{106} = 0.05 \text{ mol/dm}^3 \text{ of } \text{Na}_2\text{CO}_3 \\ \text{moles} &= 0.0125 \text{ mol} \end{aligned}$$

X